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RESEARCH INTERESTS

Protein–lipid recognition and its molecular logic; AI-driven and de novo protein design; high-throughput biophysical measurement; interpretable machine learning that turns large-scale binding data into general rules.

EDUCATION

2024.10 – present	Ph.D. Candidate, Chemistry and Biotechnology Graduate School of Engineering, The University of Tokyo · Tsuboyama Laboratory, IIS Advisor: Dr. Kotaro Tsuboyama. Large-scale mapping of protein–phosphoinositide recognition, combining high-throughput binding measurement with machine learning and AI-based protein design. Expected completion September 2027.
2024.04 – 2024.09	Research Student Institute of Industrial Science, The University of Tokyo · Tsuboyama Laboratory Joined the laboratory and established the lipid-binding screening pipeline.
2022.04 – 2024.03	M.Sc., Life Sciences Department of Interdisciplinary Sciences, Graduate School of Arts and Sciences, The University of Tokyo · Okada Laboratory (Laboratory of Pathology and Development), Institute for Quantitative Biosciences Advisor: Prof. Yuki Okada. Thesis: <i>Establishment of an Ex Vivo Sperm Chromatin Manipulation Method via Divalent Cations</i> 二価陽イオンを用いた精子クロマチンの体外改変方法の確立 · Outstanding Graduate Award (広域科学専攻奨励賞)
2018.04 – 2022.03	B.Sc., Bioscience Faculty of Life Sciences, Tokyo University of Agriculture · Laboratory of Functional Molecular Analysis (Yajima Laboratory) Advisor: Prof. Shunsuke Yajima. Thesis: <i>Structural Analysis of LgnR, an IclR-Family Transcription Factor from Paracoccus sp. 43P</i> Paracoccus sp. 43P 由来の IclR ファミリー転写因子である LgnR の構造解析
2016.10 – 2018.03	Japanese Language Program Fuji International Language Institute, Tokyo

RESEARCH POSITIONS

2024.10 – present	Research Fellow, JST SPRING-GX Support for Pioneering Research Initiated by the Next Generation, The University of Tokyo
2022.04 – 2024.03	Technical Assistant (技術補佐員) Institute for Quantitative Biosciences (IQB), The University of Tokyo

PEER-REVIEWED PUBLICATIONS

- [1] Yao, P.; Nishimura, T.*; Tsuboyama, K.* Quantitative Kinetic Analysis of Protein–Phosphoinositide Binding by Optimized Liposome-Based Bio-Layer Interferometry. *Biochemistry* **2026**, 65 (16), 2557–2565, special issue “Lipids and Lipidation”. DOI: 10.1021/acs.biochem.6c00344

Additional first-author manuscripts in preparation.

CONFERENCE PRESENTATIONS

2026.10	De novo design of phosphoinositide-binding proteins RosettaCon Asia 2026, Peking University, Beijing, China. Poster (international), upcoming.
2025.01	Decoding the Universal Principles of Protein–Phospholipid Binding <u>Yao, P.</u> ; Nishimura, T.; Tsuboyama, K. · IPR International Conference 2025, Awaji Island, Japan. Poster (international).

2023.12	Ex Vivo Sperm Chromatin Reconstitution: An Innovative Approach Institute for Quantitative Biosciences Research Exchange Meeting 2023, The University of Tokyo. Poster P-25.
2023.06	Establishment of an ex vivo Sperm Chromatin Reconstitution Method 16th Annual Meeting of the Japanese Society for Epigenetics, Hitotsubashi Hall, Tokyo. Poster P-49.
2023.06	Effects of Divalent Cations on Sperm Chromatin Structure and Ex Vivo Reconstitution 22nd UTokyo Life Sciences Symposium (BIO UT), Komaba Campus. Poster.
2022.11	Analysis of Divalent Cation Effects on Sperm Chromatin Structure Joint “Wakate-no-kai 2022”, Rinku, Osaka. Poster P-25.

GRANTS AND FELLOWSHIPS

TERM	PROGRAMME	PROJECT	OUTCOME
2024 – present	JST SPRING-GX	Support for Pioneering Research Initiated by the Next Generation: Green Transformation Program for Advanced Human Resource Development, The University of Tokyo	AWARDED
FY2027	JSPS DC2	Elucidating the precise lipid-species recognition mechanism by large-scale analysis and machine learning	UNDER REVIEW
2026	SPReAD	MEXT “AI for Science” programme: Supporting Pioneering Research through AI for 1,000 Discovery challenges, 1st call. Predicting the lipid-binding specificity of lipid-binding domains and extracting recognition rules with a sequence–structure integrated AI	PASSED REVIEW, NOT DRAWN
FY2026	JSPS DC2	Analysis of protein lifetime determinants by large-scale measurement and machine learning	NOT AWARDED
FY2025	JSPS DC1	Analysis of protein lifetime determinants by large-scale measurement and machine learning	NOT AWARDED
FY2024	JSPS DC1	Elucidating the significance of the sperm epigenome through ex vivo sperm reconstitution	NOT AWARDED

AWARDS AND ACADEMIC ACTIVITIES

2024.03	Outstanding Graduate Award, Interdisciplinary Sciences (広域科学専攻奨励賞) Graduate School of Arts and Sciences, The University of Tokyo, FY2023.
2023.08	Bio-SPM Summer School, collaborative project accepted 11th Bio-SPM Summer School, NanoLSI, Kanazawa University. High-speed AFM imaging of <i>in vitro</i> reconstituted complexes.

TECHNICAL SKILLS

Biophysics	Bio-layer interferometry (BLI) kinetics on liposome surfaces; liposome preparation of defined lipid composition; buffer optimization for protein–lipid interfaces.
High-throughput	Yeast surface display; pooled library construction and selection; next-generation sequencing and downstream analysis; fluorescence-activated cell sorting (FACS).
Protein work	Recombinant expression in <i>E. coli</i> ; affinity, ion-exchange and size-exclusion chromatography; Western blot; site-directed mutagenesis and mutant series design.
Cell & imaging	Fluorescence microscopy; intracytoplasmic sperm injection (ICSI) and mouse embryo handling; comet assay; chromatin decondensation–recondensation assays.
Computational	Structure prediction: AlphaFold2, AlphaFold3, RoseTTAFold, ESMFold2, Boltz-2, Chai-1. Generative design: RFdiffusion3, BoltzGen, NISE, DISCO, ProteinMPNN, LigandMPNN. Protein language models: ESM-2, ESM3, ESM-C, SaProt. Surface and interface analysis: MaSIF. Machine learning on large-scale sequence–function datasets.
Model organisms	<i>Escherichia coli</i> , <i>Saccharomyces cerevisiae</i> , mouse.
Languages	Chinese (native) · Japanese (advanced; full academic working proficiency, resident in Japan since 2016) · English (advanced; research, writing and presentation).